

$$01) a) \int_0^3 2xe^x dx = 4e^3 + 2 \text{ u.a.}$$

$$\int 2xe^x dx = 2 \int x \cdot e^x dx = 2 [x \cdot e^x - \int e^x \cdot 1 dx] = 2xe^x - 2e^x + C$$

$$x = u \quad dv = e^x$$

$$1 = du \quad v = e^x$$

$$\int_0^3 2xe^x dx = 2xe^x - 2e^x \Big|_0^3 = 2 \cdot (3) \cdot e^3 - 2 \cdot e^3 - (\cancel{2 \cdot (0)} \cdot e^0 - 2e^0) =$$

$$6 \cdot e^3 - 2e^3 + 2 = 4e^3 + 2$$

$$01) b) \int_1^4 \frac{x^3 + 1}{x^3} dx = \frac{111}{32} \text{ u.a.}$$

$$\int \frac{x^3}{x^3} + \frac{1}{x^3} dx = \int 1 dx + \int \frac{1}{x^3} dx = \int 1 dx + \int x^{-3} dx = x - \frac{1}{2x^2} + C$$

$$\int_1^4 \frac{x^3 + 1}{x^3} dx = x - \frac{1}{2x^2} \Big|_1^4 = 4 - \frac{1}{2 \cdot (4)^2} - 1 + \frac{1}{2 \cdot (1)^2} = \frac{96}{32} - \frac{1}{32} + \frac{16}{32} =$$

$$\frac{111}{32}$$

$$01) c) \int_1^4 \frac{2x}{\sqrt[3]{x^2-1}} dx = 3 \cdot \frac{\sqrt[3]{225}}{2} \text{ u.a.}$$

$$\int \frac{2x}{\sqrt[3]{x^2-1}} = \int \frac{1}{\sqrt[3]{u}} du = \int u^{-\frac{1}{3}} du = 3 \frac{u^{\frac{2}{3}}}{\frac{2}{3}} + C = 3 \cdot \frac{(x^2-1)^{\frac{2}{3}}}{2} + C$$

$$\begin{aligned} u &= x^2 - 1 \\ \frac{du}{dx} &= 2x \\ du &= 2x dx \end{aligned}$$

$$\int_1^4 \frac{2x}{\sqrt[3]{x^2-1}} dx = 3 \cdot \frac{(x^2-1)^{\frac{2}{3}}}{2} \Big|_1^4 = 3 \cdot \frac{((4)^2-1)^{\frac{2}{3}}}{2} - 3 \cdot \frac{((1)^2-1)^{\frac{2}{3}}}{2} =$$

$$3 \cdot \frac{(15)^{\frac{2}{3}}}{2} - 3 \cdot \frac{(0)^{\frac{2}{3}}}{2} =$$

$$3 \cdot \frac{\sqrt[3]{225}}{2}$$

$$01) d) e^{\frac{\pi}{2}} \int_1^{\frac{\pi}{2}} \frac{\text{sen}(\ln x)}{x} dx = +1 \text{ u.a.}$$

$$\int \frac{\text{sen}(\ln x)}{x} dx = \int \text{sen}(u) du = -\cos(\ln x) + C$$

$$\begin{aligned} u &= \ln x \\ \frac{du}{dx} &= \frac{1}{x} \\ du &= \frac{1}{x} dx \end{aligned}$$

$$e^{\frac{\pi}{2}} \int_1^{\frac{\pi}{2}} \frac{\text{sen}(\ln x)}{x} dx = -\cos(\ln x) \Big|_1^{\frac{\pi}{2}} + \cos(0) - \cos(\ln e^{\frac{\pi}{2}}) =$$

$$+\cos(0) - \cos\left(\frac{\pi}{2}\right) =$$

$$+1 - 0 = +1$$

$$01) e) \int_0^{\frac{\pi}{2}} \frac{\operatorname{sen}(x)}{1+\cos^2(x)} dx = \frac{\pi}{4} \text{ u.a.}$$

$$\int \frac{\operatorname{sen}(x)}{1+\cos^2(x)} dx = -\int \frac{1}{1+u^2} du = -\arctan(\cos(x)) + C$$

$$u = \cos(x)$$

$$-du = \operatorname{sen}(x) dx$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{\operatorname{sen}(x)}{1+\cos^2(x)} dx &= -\arctan(\cos(x)) \Big|_0^{\frac{\pi}{2}} = -\arctan(\cos(\frac{\pi}{2})) + \arctan(\cos(0)) \\ &= -\arctan(0) + \arctan(1) \\ &= \arctan(1) \\ &= \frac{\pi}{4} \end{aligned}$$

$$01) \oint_{-\infty}^{+\infty} \frac{8}{9+x^2} dx = \frac{8\pi}{3} \text{ u. a.}$$

$$\int \frac{8}{9+x^2} dx = 8 \int \frac{1}{3^2+x^2} dx = \frac{8}{3} \cdot \arctan\left(\frac{x}{3}\right) + C$$

$$\int_{-\infty}^{+\infty} \frac{8}{9+x^2} dx = \frac{8}{3} \cdot \arctan\left(\frac{x}{3}\right) \Big|_{-\infty}^{+\infty} = \frac{8}{3} \cdot \arctan\left(\frac{+\infty}{3}\right) - \frac{8}{3} \cdot \arctan\left(\frac{-\infty}{3}\right) =$$

$$\frac{8}{3} \left[\arctan\left(\frac{+\infty}{3}\right) - \arctan\left(\frac{-\infty}{3}\right) \right]$$

$$\frac{8}{3} \cdot \left[\frac{\pi}{2} + \frac{\pi}{2} \right]$$

$$\frac{8\pi}{3}$$

$$01) \textcircled{g} \int_1^3 \frac{x^6 - 2}{x^4} dx = \frac{650}{81} \text{ u.a.}$$

$$\int \frac{x^6 - 2}{x^4} dx = \int \frac{x^6}{x^4} dx - 2 \int \frac{1}{x^4} dx = \int x^2 dx - 2 \int x^{-4} dx = \frac{x^3}{3} + 2 \frac{1}{3x^3} + C$$

$$\int_1^3 \frac{x^6 - 2}{x^4} dx = \frac{x^3}{3} + 2 \cdot \frac{1}{3x^3} \Big|_1^3 = \frac{(3)^3}{3} + 2 \cdot \frac{1}{3 \cdot (3)^3} - \left(\frac{1^3}{3} + 2 \cdot \frac{1}{3 \cdot (1)^3} \right) =$$

$$8 + \frac{2}{81} =$$

$$\begin{array}{r} \times 81 \\ \hline 8 \\ 64 \\ \hline 648 \end{array} \quad \frac{648}{81} + \frac{2}{81} =$$

$$\frac{650}{81}$$

$$01) \textcircled{h} \int_0^{\ln 2} 4e^{-\sqrt{x}} dx = -8e^{-\sqrt{\ln 2}} + 8e^{-\sqrt{\ln 2}} - 8 \text{ u.a.}$$

$$\begin{aligned} \int 4e^{-\sqrt{x}} dx &= 4 \int e^{-\sqrt{x}} dx = 4 \int e^u \cdot (-2) \cdot u \cdot du = -8 \int u \cdot e^u du \\ \frac{du}{dx} &= -\frac{1}{2\sqrt{x}} & -8 \cdot \left[u \cdot e^u - \int e^u \cdot 1 du \right] &= \begin{matrix} t=u & dv=e^u \\ dt=1 & v=e^u \end{matrix} \\ -2\sqrt{x} du &= dx & -8 \cdot \left[u \cdot e^u - e^u \right] &= \\ & & -8e^u u + 8e^u + C &= \\ & & -8e^{-\sqrt{x}} (-\sqrt{x}) + 8e^{-\sqrt{x}} + C & \end{aligned}$$

$$\begin{aligned} \int_0^{\ln 2} 4e^{-\sqrt{x}} dx &= -8e^{-\sqrt{x}} (-\sqrt{x}) + 8e^{-\sqrt{x}} \Big|_0^{\ln 2} = -8e^{-\sqrt{\ln 2}} + 8e^{-\sqrt{\ln 2}} - \left(-8e^{-\sqrt{0}} + 8e^{-\sqrt{0}} \right) \\ &= -8e^{-\sqrt{\ln 2}} + 8e^{-\sqrt{\ln 2}} - 8 \end{aligned}$$

$$01) i) \int_{-2}^{54} x^3(6x-8)dx = 6 \cdot \frac{54^5}{5} - 8 \cdot \frac{54^4}{4} - 6 \cdot \frac{-2^5}{5} + 8 \cdot \frac{-2^4}{4} \text{ u.a.}$$

$$\int x^3(6x-8)dx = \int 6x^4 - 8x^3 dx = 6 \int x^4 dx - 8 \int x^3 dx = 6 \cdot \frac{x^5}{5} - 8 \cdot \frac{x^4}{4} + C$$

$$\int_{-2}^{54} x^3(6x-8) = 6 \cdot \frac{x^5}{5} - 8 \cdot \frac{x^4}{4} \Big|_{-2}^{54} = 6 \cdot \frac{54^5}{5} - 8 \cdot \frac{54^4}{4} - 6 \cdot \frac{-2^5}{5} + 8 \cdot \frac{-2^4}{4}$$

$$01) ii) \int_{-2}^6 x^3(4x^2-8)dx = \frac{85504}{3} \text{ u.a.}$$

$$\int x^3(4x^2-8)dx = \int 4x^5 - 8x^3 dx = 4 \int x^5 dx - 8 \int x^3 dx =$$

$$\int_{-2}^6 x^3(4x^2-8)dx = 4 \cdot \frac{x^6}{6} - 8 \cdot \frac{x^4}{4} = \frac{4 \cdot x^6}{6} - 8 \cdot \frac{x^4}{4} + C$$

$$\frac{4 \cdot x^6}{6} - 8 \cdot \frac{x^4}{4} \Big|_{-2}^6 = 4 \cdot \frac{6^6}{6} - 8 \cdot \frac{6^4}{4} - 4 \cdot \frac{(-2)^6}{6} + 8 \cdot \frac{(-2)^4}{4}$$

$$\frac{85504}{3}$$

01) (K) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^3 x \cdot \sin x \, dx = + \frac{\cos^4(\frac{\pi}{4})}{4} \text{ u.a.}$

$$\int \cos^3 x \cdot \sin x \, dx = - \int u^3 \, du = - \frac{\cos^4(x)}{4} + C \quad \begin{array}{l} u = \cos x \\ du = -\sin x \, dx \end{array}$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^3(x) \cdot \sin(x) \, dx = - \frac{\cos^4(x)}{4} \bigg|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = - \frac{\cancel{\cos^4(\frac{\pi}{2})}}{4} + \frac{\cos^4(\frac{\pi}{4})}{4} = + \frac{\cos^4(\frac{\pi}{4})}{4}$$

01) (L) $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^3 x \cdot \cos x \, dx = \frac{1}{4} - \frac{\sin^4(\frac{\pi}{4})}{4} \text{ u.a.}$

$$\int \sin^3 x \cdot \cos x \, dx = \int u^3 \, du = \frac{\sin^4(x)}{4} + C \quad \begin{array}{l} u = \sin x \\ du = \cos x \, dx \end{array}$$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^3(x) \cdot \cos(x) \, dx = \frac{\sin^4(x)}{4} \bigg|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = \frac{\sin^4(\frac{\pi}{2})}{4} - \frac{\sin^4(\frac{\pi}{4})}{4} = \frac{1}{4} - \frac{\sin^4(\frac{\pi}{4})}{4}$$

$$01) \textcircled{m} \int \sin^5(x) dx = \int (1 - \cos^2 x)^2 \cdot \sin(x) dx =$$

$$\begin{array}{l} u = \cos x \\ du = -\sin x dx \end{array} \quad - \int (1 - u^2)^2 du = - \int 1 - 2u^2 + u^4 du =$$

$$\begin{array}{l} (1-u^2)(1-u^2) \\ 1-u^2-u^2+u^4 \end{array} \quad - \int 1 du + 2 \int u^2 du - \int u^4 du = -u + 2 \frac{u^3}{3} - \frac{u^5}{5}$$

$$- \cos(x) + 2 \frac{\cos^3(x)}{3} - \frac{\cos^5(x)}{5} + C$$

$$01) \textcircled{n} \int \cos^5(x) dx = \int (1 - \sin^2(x))^2 \cdot \cos(x) dx =$$

$$\begin{array}{l} u = \sin(x) \\ du = \cos(x) dx \end{array} \quad \int (1 - u^2)^2 \cdot du$$

$$\int 1 - 2u^2 + u^4 du =$$

$$\int 1 du - 2 \int u^2 + \int u^4 du =$$

$$u - 2 \frac{u^3}{3} + \frac{u^5}{5} + C$$

$$\sin(x) - 2 \frac{\sin^3(x)}{3} + \frac{\sin^5(x)}{5} + C$$

$$\begin{aligned}
 01)_{\text{a}} \int \operatorname{tg}^5(x) dx &= \frac{\operatorname{tg}^4(x)}{4} - \int \operatorname{tg}^3(x) dx = \\
 &= \frac{\operatorname{tg}^4(x)}{4} - \left(\frac{\operatorname{tg}^2(x)}{2} - \int \operatorname{tg}(x) dx \right) \\
 &= \frac{\operatorname{tg}^4(x)}{4} - \left(\frac{\operatorname{tg}^2(x)}{2} - \int \frac{\operatorname{sen}(x)}{\cos(x)} dx \right) \\
 &\quad \begin{array}{l} u = \cos(x) \\ -du = \operatorname{sen}(x) dx \end{array} \\
 &= \frac{\operatorname{tg}^4(x)}{4} - \left(\frac{\operatorname{tg}^2(x)}{2} + \int \frac{1}{u} du \right) \\
 &= \frac{\operatorname{tg}^4(x)}{4} - \frac{\operatorname{tg}^2(x)}{2} - \ln|\cos(x)| + C
 \end{aligned}$$

$$\begin{aligned}
 01)_{\text{b}} \int \operatorname{cotg}^5(x) dx &= -\frac{\operatorname{cotg}^4(x)}{4} - \int \operatorname{cotg}^3(x) dx = \\
 &= -\frac{\operatorname{cotg}^4(x)}{4} - \left(-\frac{\operatorname{cotg}^2(x)}{2} - \int \frac{\operatorname{sen}(x)}{\cos(x)} dx \right) \\
 &\quad \begin{array}{l} u = \cos(x) \\ -du = \operatorname{sen}(x) dx \end{array} \\
 &= -\frac{\operatorname{cotg}^4(x)}{4} - \left(-\frac{\operatorname{cotg}^2(x)}{2} + \int \frac{1}{u} du \right) \\
 &= -\frac{\operatorname{cotg}^4(x)}{4} + \frac{\operatorname{cotg}^2(x)}{2} + \ln|\operatorname{sen}(x)| + C
 \end{aligned}$$

$$\begin{aligned}
 \text{O1) } \textcircled{P} \int \sec^5 x \, dx &= \frac{\sec^3 x \operatorname{tg} x}{4} + \frac{3}{4} \int \sec^3 x \, dx = \\
 &\frac{\sec^3 x \operatorname{tg} x}{4} + \frac{3}{4} \left[\frac{\sec x \operatorname{tg} x}{2} + \frac{1}{2} \int \sec x \, dx \right] \\
 &\frac{\sec^3 x \cdot \operatorname{tg} x}{4} + \frac{3}{4} \left[\frac{\sec x \operatorname{tg} x}{2} + \frac{1}{2} \ln |\sec x + \operatorname{tg} x| \right] + C \\
 &\frac{\sec^3 x \cdot \operatorname{tg} x}{4} + \frac{3 \sec x \cdot \operatorname{tg} x}{8} + \frac{3 \ln |\sec x + \operatorname{tg} x|}{8} + C
 \end{aligned}$$

$$\begin{aligned}
 \text{O1) } \textcircled{P} \int \cos \sec^5 x \, dx &= -\frac{\cos \sec^3 x \cdot \cot x}{4} + \frac{3}{4} \int \cos \sec^3 x \, dx \\
 &-\frac{\cos \sec^3 x \cdot \cot x}{4} + \frac{3}{4} \left[-\frac{\cos \sec x \cdot \cot x}{2} + \frac{1}{2} \int \cos \sec x \, dx \right] \\
 &-\frac{\cos \sec^3 x \cdot \cot x}{4} + \frac{3}{4} \left[-\frac{\cos \sec x \cdot \cot x}{2} - \frac{\ln |\cos \sec x + \cot x|}{2} \right] \\
 &-\frac{\cos \sec^3 x \cdot \cot x}{4} - \frac{3 \cos \sec x \cdot \cot x}{8} + \frac{3 \ln |\cos \sec x + \cot x|}{8} + C
 \end{aligned}$$

$$01) \textcircled{1} \int \cos^4 x \cdot \sin^4 x \, dx$$

$$\int \left(\frac{1 + \cos(2x)}{2} \right)^2 \cdot \left(\frac{1 - \cos(2x)}{2} \right)^2 dx$$

$$\int \left(\frac{1 + \cos^2(2x)}{4} \right) \cdot \left(\frac{1 - \cos^2(2x)}{4} \right) dx$$

$$\int \frac{1 - \cos^4(2x)}{16} dx$$

$$\int \frac{\sin^4(2x)}{16} dx$$

$$\frac{1}{16} \int \sin^4(2x) dx \quad \begin{matrix} u = 2x \\ du = 2 dx \end{matrix}$$

$$\frac{1}{32} \int \sin^4 u \, du = \frac{1}{32} \left[-\frac{\sin^3 x \cdot \cos x}{4} + \frac{3}{4} \int \sin^2 x \, dx \right] =$$

$$\frac{1}{32} \left[-\frac{\sin^3 x \cdot \cos x}{4} + \frac{3}{4} \left[-\frac{\sin x \cdot \cos x}{2} + \frac{1}{2} \int 1 \, dx \right] \right]$$

$$\frac{1}{32} \left[-\frac{\sin x \cdot \cos x}{4} + \frac{3}{4} \left[-\frac{\sin x \cdot \cos x}{2} + \frac{x}{2} \right] \right] + C$$

$$01) 2) \int \cos^6 x \sin^4 x dx = \int \left(\frac{1+\cos(2x)}{2} \right)^3 \left(\frac{1-\cos^2(2x)}{2} \right)^2 dx$$

$$\int \frac{(1+\cos(2x))^3}{2^3} \cdot \frac{(1-\cos(2x))^2}{2^2} dx$$

$$\frac{1}{32} \int (1+\cos(2x))^3 \cdot (1-\cos(2x))^2 dx \quad u = \cos(2x)$$

$$\frac{1}{32} \int (1+u)^3 (1-u)^2 dx =$$

$$\frac{1}{32} \int (1+3u+3u^2+u^3)(1-2u+u^2) dx$$

$$\frac{1}{32} \int \cancel{1-2u+u^2} + \cancel{3u-6u^2+3u^3} + \cancel{3u^2-6u^3+3u^4} + \cancel{u^3-2u^4+u^5} dx$$

$$\frac{1}{32} \int 1+u-2u^2-2u^3+u^4+u^5 dx$$

$$\frac{1}{32} \left[\int 1 dx + \int \cos(2x) dx - 2 \int \cos^2(2x) dx - 2 \int \cos^3(2x) dx + \int \cos^4(2x) dx + \int \cos^5(2x) dx \right]$$

$$01) ③ \int \cos^7 x \cdot \sin^4 x \, dx$$

$$\int \sin^4 x \cdot \cos^6 x \cdot \cos x \, dx$$

$$\int \sin^4 x \cdot (1 - \sin^2 x) \cdot (1 - \sin^2 x) \cdot (1 - \sin^2 x) \cdot \cos x \, dx$$

$$\begin{array}{l} u = \sin x \\ du = \cos x \, dx \end{array} \int u^4 \cdot (1 - u^2) \cdot (1 - u^2) \cdot (1 - u^2) \, du$$

$$\int u^4 \cdot (1 - 3u^2 + 3u^4 - u^6) \, du$$

$$\int u^4 - 3u^6 + 3u^8 - u^{10} \, du$$

$$\frac{u^5}{5} - 3\frac{u^7}{7} + 3\frac{u^9}{9} - \frac{u^{11}}{11} + C$$

$$\frac{\sin^5 x}{5} - \frac{3\sin^7 x}{7} + \frac{3\sin^9 x}{9} - \frac{\sin^{11} x}{11} + C$$

$$01) \textcircled{4} \int \cos^4 x \sin^3 x \, dx$$

$$\int \cos^4 x \cdot \sin^2 x \cdot \sin x \, dx$$

$$\int \cos^4 x \cdot (1 - \cos^2 x) \cdot \sin x \, dx$$

$$u = \cos x$$

$$-du = \sin x \cdot dx$$

$$-\int u^4 \cdot (1 - u^2) \, du$$

$$-\int u^4 - u^6 \, du$$

$$-\frac{u^5}{5} + \frac{u^7}{7} + C$$

$$-\frac{\cos^5 x}{5} + \frac{\cos^7 x}{7} + C$$

$$01) ⑤ \int \tan^9 x \cdot \sec^6 x \, dx$$

$$\int \tan^9 x \cdot (1 + \tan^2 x)^2 \cdot \sec^2 x \, dx$$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$\int u^9 \cdot (1 + u^2)^2 \, du$$

$$\int u^9 \cdot (1 + 2u^2 + u^4) \, du$$

$$\int u^9 + 2u^{11} + u^{13} \, du$$

$$\frac{u^{10}}{10} + 2 \frac{u^{12}}{12} + \frac{u^{14}}{14} + C$$

$$\frac{\tan^{10} x}{10} + 2 \frac{\tan^{12} x}{12} + \frac{\tan^{14} x}{14} + C$$

$$01) ⑥ \int \operatorname{tg} x \cdot \sec^7 x \, dx$$

$$\int (-1 + \sec^2 x)^5 \cdot \operatorname{tg} x \cdot \sec x \cdot \sec^6 x \, dx$$

$$\begin{aligned} u &= \sec x \\ du &= \sec x \cdot \operatorname{tg} x \, dx \end{aligned} \quad \int (-1 + u^2)^5 \cdot u^6 \, du$$

$$\int u^{16} - 5u^{14} + 10u^{12} - 10u^{10} + 5u^8 - u^6 \, du$$

$$\frac{\sec^{17} x}{17} - \frac{5 \sec^{15} x}{15} + \frac{10 \sec^{13} x}{13} - \frac{10 \sec^{11} x}{11} + \frac{5 \sec^9 x}{9} - \frac{\sec^7 x}{7} + C$$

$$Q1) 08) \int \frac{-4x^3}{2x^3 + x^2 - 2x - 1} dx =$$

$$\int \frac{-4x^3 + 6x^3 + x^2 - 2x - 1 - 6x^3 - x^2 + 2x + 1}{2x^3 + x^2 - 2x - 1} dx$$

$$\int \frac{2x^3 + x^2 - 2x - 1}{2x^3 + x^2 - 2x - 1} dx + \int \frac{-6x^3 - x^2 + 2x + 1}{2x^3 + x^2 - 2x - 1} dx$$

$$\int 1 dx + \int \frac{-6x^3 - x^2 + 2x + 1}{2x^3 + x^2 - 2x - 1} dx$$

$$C + x - \int 3 dx - \frac{1}{3} \int \frac{1}{2x+1} dx - \frac{2}{3} \int \frac{1}{x-1} dx + 2 \int \frac{1}{x+1} dx$$

$$-2x - \frac{1}{3} \cdot \ln|2x+1| - \frac{2}{3} \cdot \ln|x-1| + 2|x+1| + C$$

01) 15) $\int_{-\infty}^{+\infty} e^{x^2} dx$

$\int e^{x^2} dx = \text{NÃO TEM PRIMITIVA}$

Logo $\int_{-\infty}^{+\infty} e^{x^2} dx$ cresce mais rápido do que qualquer função e não tem área finita sob ela.

$$02) y = 2x \quad y = 2x$$

$$x^2 = 2x \quad \left| \int_0^2 x^2 - 2x dx \right| =$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0 \quad \left| \int_0^2 x^2 dx - \int_0^2 2x dx \right| =$$

$$x = 0$$

$$\left| \frac{x^3}{3} \right|_0^2 \left| \frac{2^3}{3} - \frac{0^3}{3} - x^2 \right|_0^2 \left| 2^2 - 0^2 \right| =$$

or

$$x = 2$$

$$\left| \frac{8}{3} - 4 \right| =$$

$$\left| \frac{8}{3} - \frac{12}{3} \right| =$$

$$\left| -\frac{4}{3} \right| = \frac{4}{3} \text{ u.a.}$$

$$03) y = x^3 \quad y = 4x$$

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

$$x = 0 \text{ ou } x = 2 \text{ ou } x = -2$$

$$[-2, 2] 2 \left| \int_0^2 4x - x^3 dx \right|$$

$$2 \left[\int_0^2 4x dx - \int_0^2 x^3 dx \right]$$

$$2 \left[\left[2x^2 \right]_0^2 - \left[\frac{x^4}{4} \right]_0^2 \right]$$

$$2 [8 - 4] = 8 \text{ u.a.}$$

$$04) y = x^2 - 1 \quad y = x + 1$$

$$x^2 - 1 = x + 1 \quad [-1, 2]$$

$$x^2 - x - 2 = 0 \quad \Delta = 1 - 4 \cdot (-2) \cdot 1 = 9$$

$$\int_{-1}^2 (x^2 - 1) - (x + 1) dx$$

$$x_1 = \frac{1+3}{2} = \frac{4}{2} = 2 \quad \frac{x^3}{3} - \frac{x^2}{2} - 2x \quad \left| \frac{2^3}{3} - \frac{2^2}{2} - 2 \cdot 2 - \frac{(-1)^3}{3} + \frac{(-1)^2}{2} + 2 \cdot (-1) \right|$$

$$x_2 = \frac{1-3}{2} = \frac{-2}{2} = -1$$

$$\left| \frac{8}{3} - 2 - 4 + \frac{1}{3} + \frac{1}{2} - 2 \right|$$

$$\left| \frac{8}{3} - 8 + \frac{1}{3} + \frac{1}{2} \right|$$

$$\left| \frac{9}{3} - 8 + \frac{1}{2} \right|$$

$$\left| 3 - 8 + \frac{1}{2} \right|$$

$$\left| -5 + \frac{1}{2} \right|$$

$$\left| \frac{-10}{2} + \frac{1}{2} \right| = \frac{9}{2} \text{ m.a.}$$

$$05) Y = x^2 + 1 \quad Y = 3x + 1$$

$$x^2 + 1 = 3x + 1 \quad [0, 3]$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$x = 0 \text{ ou } x = 3$$

$$\int_0^3 (x^2 + 1) - (3x + 1) dx$$

$$\int_0^3 x^2 + 1 - 3x - 1 dx$$

$$\int_0^3 x^2 - 3x dx$$

$$\frac{x^3}{3} - 3\frac{x^2}{2} \Big|_0^3 = \frac{3^3}{3} - 3\frac{3^2}{2} - \frac{0^3}{3} + 3\frac{0^2}{2}$$

$$\left| \frac{27}{3} - 3\frac{9}{2} \right|$$

$$\left| \frac{27}{3} - \frac{27}{2} \right|$$

$$\left| \frac{54}{6} - \frac{81}{6} \right|$$

$$\left| \frac{-27}{6} \right| = \frac{27}{6} = \frac{9}{2} \text{ u.a.}$$

06) $y = x + 6$ $y = x^3$ $y = -\frac{x}{2}$

$x + 6 = x^3$
 $x^3 - x - 6 = 0$

$x = 2 \leftarrow \text{Raíz}$
 // ————— //

$x + 6 = -\frac{x}{2}$
 $2x + 12 = -x$
 $3x + 12 = 0$
 $x + 4 = 0$
 $x = -4 \leftarrow \text{Raíz}$
 // ————— //

$x^3 = -\frac{x}{2}$
 $2x^3 = -x$
 $2x^3 + x = 0$
 $x(2x^2 + 1) = 0$
 $x = 0 \leftarrow \text{Raíz}$

A_1 $[-4, 0] \cup [0, 2]$ A_2

$\left| \int_{-4}^0 (x+6) - \left(-\frac{x}{2}\right) dx \right| =$
 $\left| \int_{-4}^0 x + 6 + \frac{x}{2} dx \right| =$
 $\left| \int_{-4}^0 x dx + \int_{-4}^0 6 dx + \frac{1}{2} \int_{-4}^0 x dx \right| =$
 $\left| \frac{x^2}{2} + 6x + \frac{x^2}{4} \right|_{-4}^0 =$
 $\left| 0 - \frac{(-4)^2}{2} - 6 \cdot (-4) - \frac{(-4)^2}{4} \right| =$
 $|-8 + 24 - 4| = 12 = A_1$

$\left| \int_0^2 x^3 - x - 6 dx \right| =$
 $\left| \frac{x^4}{4} - \frac{x^2}{2} - 6x \right|_0^2 = \left| \frac{2^4}{4} - \frac{2^2}{2} - 6 \cdot 2 \right| =$
 $|4 - 2 - 12| = 10$
 A_2

$$A_1 + A_2 = A_T$$

$$12 + 10 = 22 \text{ u.a.}$$